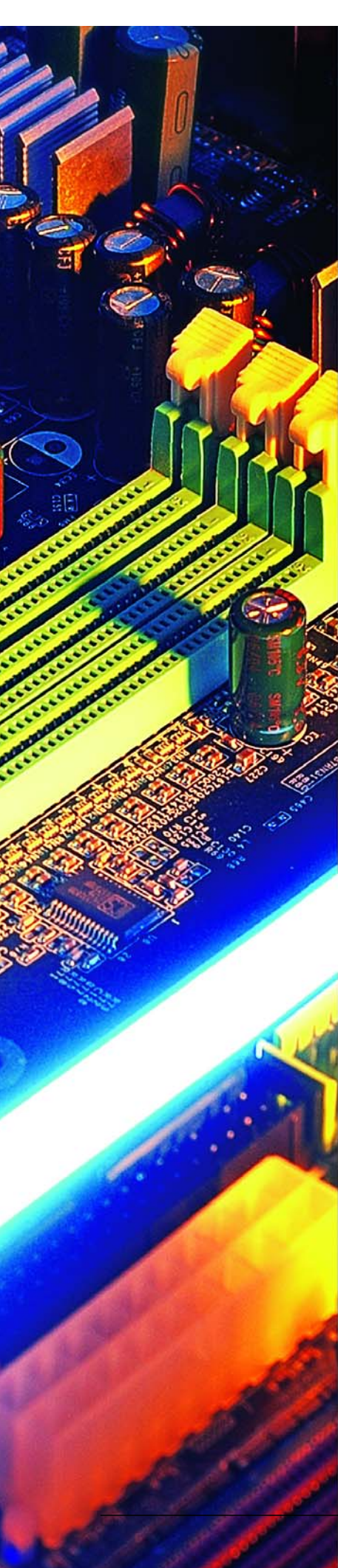




FUTURE PERFECT

The complete guide to choosing the best platform that is easy on the wallet, and will last you for years

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Evolution on Earth is a lethargic process. Events in the PC market follow a similar timeline, with innovations moving the industry at a leisurely pace. It is thus quite a sight to witness the barren lands of the PC—wrecked by bandwidth limitations and littered with GHz myths—sprouting next-generation technologies that promise to turn the desert into thriving wetlands. Sleek and efficient dual-core processors replacing fire-breathing, single-core GHz monsters is a good thing. Bottlenecks are, similarly, being eradicated by the replacing of parallel connectors with serial ones—witness the PCI-Express bus and SATA hard disks. Graphics cards now sport CPU-like processors which will soon be better utilised by operating systems across the board, bringing performance and usability to an ageing GUI.

The PC industry is shifting gears, and not a moment too soon. This change in pace brings along new jargon to sift through. New chipsets, new processor roadmaps, new memory technologies and the like take on greater significance. Change brings along confusion; our first task here will be to shed light on the scene.

Our goal here is not to choose the best processor and the best motherboard. Rather, it is to pick the best combination. We will help you arrive at a system-wide solution—a solution that will last you through a few years and that will not strain your pocket: the conclusion to this test will thus choose a combination processor/motherboard that strikes the perfect balance between performance, features and price.

With that in mind, let us first take a look at where things stand in the rival camps of industry movers Intel and AMD.

Last year, Intel introduced its 915 and 925 chipsets alongside the new LGA 775 socket processors, which were meant to be paired with these chipsets. The highlight of these chipsets was PCI-Express, offering significantly higher bandwidth than the current PCI bus (*For a complete low-down on PCI-Express, read our In Sight story on page 32*).

nVidia and ATi were the first to launch products for the new bus, and the rest of the industry will follow soon.

AMD, after the introduction of their Athlon 64 processors, are riding the waves of success. Not only are their CPUs currently at the top of the speed chart, but they have managed to rein in the heating issues that plagued them.

While Intel's product portfolio is focused and simple to understand, AMD's is a complete mess. The Athlon 64 processors are currently available in two different socket versions: Socket 939 and Socket 754. The Athlon XP range of processors recently got a replacement in the form of Sempron.

While that was on the processor side, the number of chipsets available for an AMD processor is numerous, and deciding upon one is not an easy task. This will be handled later on in this article when we look at motherboards from different manufacturers sporting various chipsets.

We often receive letters asking what kind of machine will be decent enough to play most of today's games, will offer a fair amount of upgradeability, and most importantly, will be easy enough on the wallet. Since processors and motherboards are the two most crucial components that decide the overall performance of a system, while dictating the headroom for future expansion, we decided to do a CPU and motherboard comparison test. This article will help you decide the right platform to upgrade to, or settle for, when buying a new PC.

Due to the aforementioned changes in technology, choices are many, and at this juncture choosing the right solution matters: one wrong turn and your upgrade options could hit a dead-end. So, as we compare the different processors and motherboards, we will also discuss the pros and cons of the respective chipsets and processor families.

This article is broadly classified into two parts: processors and motherboards. First, we take a look at the available processors, and then benchmark them to arrive at a winner. Similar treatment will then be applied to motherboards. We will then pick the best combination of processor